

1 Conditional Expectation

Let Y denote a real-valued random variable and $X \in \mathbb{R}^k$ denote a random vector. Throughout this section, assume that $\mathbb{E}|Y| < \infty$.

Definition 1 (Conditional Expectation). Let X and Y be two random variables whose joint density $f_{X,Y}(x,y)$ exists. The conditional expectation of Y given $X = x$ is defined as

$$\mathbb{E}(Y | X = x) := \int_{\mathbb{R}} y f_{Y|X}(y | x) dy, \quad (1)$$

where $f_{Y|X}(y | x)$ is the conditional density of Y given $X = x$.

Theorem 1 (Law of Iterated Expectations (LIE)). Let X , Y , and Z be random variables with $\mathbb{E}|Y| < \infty$. Then, we have

$$\mathbb{E}(Y | Z) = \mathbb{E}[\mathbb{E}(Y | X, Z) | Z]. \quad (2)$$

Sketch of Proof. We show the case where the joint density $f_{X,Y,Z}(x,y,z)$ exists. For any z , we have

$$\begin{aligned} \mathbb{E}[\mathbb{E}(Y | X, Z) | Z = z] &= \int \left(\int y f_{Y|X,Z}(y | x, z) dy \right) f_{X|Z}(x | z) dx \\ &= \int y \left(\int f_{Y,X|Z}(y, x | z) dx \right) dy \\ &= \int y f_{Y|Z}(y | z) dy = \mathbb{E}(Y | Z = z). \quad \blacksquare \end{aligned}$$

Remark 1. The LIE is also known as the “law of *total* expectations.”

Remark 2. In the special case where Z is a constant, the LIE simplifies to

$$\mathbb{E}(Y) = \mathbb{E}[\mathbb{E}(Y | X)]. \quad (3)$$

Theorem 2 (Conditioning Theorem). Let X and Y be two random variables with $\mathbb{E}|Y| < \infty$. Then, for any function g of X with $\mathbb{E}|g(X)Y| < \infty$, we have

$$\mathbb{E}(g(X)Y | X) = g(X) \mathbb{E}(Y | X). \quad (4)$$

Sketch of Proof. Once X is *given* or *fixed*, $g(X)$ is no longer random, so it can be treated as a constant and pulled out of the conditional expectation. $\blacksquare$

Theorem 3 (Characterization of Conditional Expectation). Suppose that $\mathbb{E}(Y^2) < \infty$. Then, $\mathbb{E}(Y | X)$ is the unique function of X that minimizes the mean-squared error loss:

$$\mathbb{E}(Y | X) = \arg \min_{g(X)} \mathbb{E}[(Y - g(X))^2]. \quad (5)$$

Here we compare predictors with finite second moments.

Proof. For simplicity, write $m(X) := \mathbb{E}(Y | X)$. Let $g(X)$ be any function of X with $\mathbb{E}[g(X)^2] < \infty$. Then we can decompose the mean squared error as follows:

$$\begin{aligned} \mathbb{E}[(Y - g(X))^2] &= \mathbb{E}[(Y - m(X) + m(X) - g(X))^2] \\ &= \mathbb{E}[(Y - m(X))^2] + \mathbb{E}[(m(X) - g(X))^2] \\ &\quad + 2 \mathbb{E}[(Y - m(X))(m(X) - g(X))]. \end{aligned}$$

By **LIE** and **Conditioning Theorem**, we have

$$\begin{aligned} \mathbb{E}[(Y - m(X))(m(X) - g(X))] &= \mathbb{E}[\mathbb{E}[(Y - m(X))(m(X) - g(X)) | X]] \\ &= \mathbb{E}[(m(X) - g(X)) \mathbb{E}[Y - m(X) | X]] = 0. \end{aligned}$$

Therefore, we have

$$\mathbb{E}[(Y - g(X))^2] = \mathbb{E}[(Y - m(X))^2] + \mathbb{E}[(m(X) - g(X))^2]. \quad (6)$$

The first term does not depend on $g(X)$. The second term is non-negative and equals zero if and only if $g(X) = m(X)$ almost surely. Hence, $m(X)$ is the unique function of X that minimizes the mean squared error. ■

Remark 3 (“Unique”). Technically, conditional expectations are unique only up to almost-sure equality. This means that two versions may differ on a set of values of X that has probability zero. That said, in this class, we will not focus on these measure-theoretic technicalities.

2 Population Regression Problem

We now let Y be a real-valued outcome and $X = (X_1, \dots, X_k)'$ be a k -dimensional vector of regressors. The properties of conditional expectations above also apply when we condition on a vector. Throughout this section, assume that $\mathbb{E}(Y^2) < \infty$ and $\mathbb{E}(X_j^2) < \infty$ for every j . When an intercept is included, the first component of X is the constant 1.

Definition 2 (Conditional Mean Decomposition). Let $m(X) := \mathbb{E}(Y | X)$ and define $\varepsilon := Y - m(X)$. Then we can write

$$Y = m(X) + \varepsilon \quad \text{with} \quad \mathbb{E}(\varepsilon | X) = 0. \quad (7)$$

Remark 4. The zero conditional mean follows directly from

$$\mathbb{E}(\varepsilon | X) = \mathbb{E}(Y - m(X) | X) = \mathbb{E}(Y | X) - m(X) = 0. \quad (8)$$

Thus, this decomposition is available for any Y and X satisfying our moment assumptions. It does not require the conditional mean to be linear.

Proposition 1 (Orthogonality of the Conditional Mean Error). For any measurable function $g(X)$ with $\mathbb{E}[g(X)^2] < \infty$, we have

$$\mathbb{E}[g(X)\varepsilon] = 0. \quad (9)$$

In particular, $\mathbb{E}(\varepsilon) = 0$ and $\mathbb{E}(X\varepsilon) = 0$.

Proof. By **LIE** and **Conditioning Theorem**,

$$\mathbb{E}[g(X)\varepsilon] = \mathbb{E}[\mathbb{E}(g(X)\varepsilon | X)] = \mathbb{E}[g(X) \mathbb{E}(\varepsilon | X)] = 0. \quad \blacksquare$$

Remark 5 (Interpretation of **Proposition 1**). The conditional mean error is orthogonal not only to each regressor, but also to every function of X with a finite second moment. Thus, no such function can be added to $m(X)$ to improve its mean squared prediction error. This does not mean that ε and X are independent: for example, the variance of ε may still depend on X .

Definition 3 (Best Linear Predictor). The population least squares coefficient is a solution to

$$\beta \in \arg \min_{b \in \mathbb{R}^k} \mathbb{E}[(Y - X'b)^2]. \quad (10)$$

The function $X'\beta$ is called the *best linear predictor* of Y given X , and $u := Y - X'\beta$ is the *population regression error*.

Remark 6. By **Characterization of Conditional Expectation**, $m(X)$ is the unique best predictor of Y under mean squared error loss. By imposing linearity as in (10), we have the following decomposition given any $b \in \mathbb{R}^k$:

$$\mathbb{E}[(Y - X'b)^2] = \mathbb{E}(\varepsilon^2) + \mathbb{E}[(m(X) - X'b)^2]. \quad (11)$$

The first term is the prediction error that remains even if we know the conditional mean. The second term measures how well $X'b$ approximates the conditional mean and is zero exactly when $m(X) = X'b$.

Theorem 4 (Population Least Squares). Suppose that $\mathbb{E}(XX')$ is nonsingular. Then the population least squares coefficient is unique and satisfies

$$\beta = \mathbb{E}(XX')^{-1} \mathbb{E}(XY). \quad (12)$$

Equivalently, it is the unique solution to the equations

$$\mathbb{E}[X(Y - X'\beta)] = 0. \quad (13)$$

Proof. Expanding the objective gives

$$\mathbb{E}[(Y - X'b)^2] = \mathbb{E}(Y^2) - 2b' \mathbb{E}(XY) + b' \mathbb{E}(XX')b.$$

Differentiating with respect to b gives the first-order condition

$$-2 \mathbb{E}(XY) + 2 \mathbb{E}(XX')b = 0, \quad (14)$$

which rearranges to the population normal equation $\mathbb{E}[X(Y - X'b)] = 0$. Since $\mathbb{E}(XX')$ is nonsingular, we have

$$\beta = \mathbb{E}(XX')^{-1} \mathbb{E}(XY). \quad (15)$$

Moreover, $\mathbb{E}(XX')$ being strictly positive definite implies that the objective is strictly convex and the first-order condition determines its unique minimizer. ■

Remark 7 (Moment Conditions). Sometimes you will see (13) referred to as the *moment conditions*. The term *moment conditions* emphasizes that they restrict population expectations.

Definition 4 (Perfect Multicollinearity). We say that X suffers from *perfect multicollinearity* if there exists a nonzero vector $v \in \mathbb{R}^k$ such that $X'v = 0$.

Lemma 1. If X does not suffer from perfect multicollinearity, then $\mathbb{E}(XX')$ is strictly positive definite.

Proof. Let $v \in \mathbb{R}^k$ be any nonzero vector. Then we have

$$v' \mathbb{E}(XX')v = \mathbb{E}(v'XX'v) = \mathbb{E}[(X'v)^2] \geq 0. \quad (16)$$

The expectation of a squared random variable can be zero only when that random variable equals zero. Without perfect multicollinearity, $X'v = 0$ is ruled out for every nonzero v , so $\mathbb{E}[(X'v)^2] > 0$. Hence, $\mathbb{E}(XX') \succ 0$. ■

Remark 8 (Identification and Perfect Multicollinearity). By **Lemma 1** and **Population Least Squares**, the population least squares coefficient β is uniquely determined when there is no perfect multicollinearity. Conversely, if $X'v = 0$ for some nonzero v , then β and $\beta + v$ give the same predictor and hence the same loss. Thus, β is identified if and only if there is no perfect multicollinearity. Perfect multicollinearity does not prevent from fitting the linear model, but it prevents us from *identifying* the coefficient.

Remark 9 (Orthogonality versus Zero Conditional Mean). **Equation 13** gives the population orthogonality condition $\mathbb{E}(Xu) = 0$, which does not in general imply $\mathbb{E}(u | X) = 0$. Notice that

$$u = \varepsilon + m(X) - X'\beta, \quad \mathbb{E}(u | X) = m(X) - X'\beta. \quad (17)$$

Thus, the regression error has zero conditional mean if and only if the conditional mean is linear in the included regressors.

Proposition 2. If an intercept is included in (10), then $\mathbb{E}(u) = 0$.

Proof. Suppose that $X_1 = 1$. Since $\mathbb{E}(Xu) = 0$, we have $\mathbb{E}(X_1u) = \mathbb{E}(u) = 0$. then the first component of (13) gives $\mathbb{E}(X_1u) = \mathbb{E}(u) = 0$. ■

Remark 10 (Intuition of Including an Intercept). An intercept allows us to shift the predictor up or down without changing its slopes. If a predictor's error u has a nonzero mean, shifting the predictor upward by $\mathbb{E}(u)$ gives the new error $u - \mathbb{E}(u)$, with strictly smaller mean squared error:

$$\mathbb{E}[(u - \mathbb{E}(u))^2] = \mathbb{E}(u^2) - \mathbb{E}(u)^2 < \mathbb{E}(u^2).$$

Thus, at the least squares optimum, there can be no remaining average overprediction or underprediction: $\mathbb{E}(u) = 0$, or equivalently, $\mathbb{E}(X'\beta) = \mathbb{E}(Y)$.

Example 1 (One Regressor and an Intercept). Let $X = (1, X_2)'$ with $\text{Var}(X_2) > 0$, and write $\beta = (\beta_0, \beta_1)'$, so that $X'\beta = \beta_0 + \beta_1 X_2$. The normal equations become

$$\begin{aligned}\mathbb{E}(Y - \beta_0 - \beta_1 X_2) &= 0, \\ \mathbb{E}[X_2(Y - \beta_0 - \beta_1 X_2)] &= 0.\end{aligned}$$

The first equation gives $\beta_0 = \mathbb{E}(Y) - \beta_1 \mathbb{E}(X_2)$. Substituting into the second equation yields

$$\beta_1 = \frac{\text{Cov}(X_2, Y)}{\text{Var}(X_2)}, \quad \beta_0 = \mathbb{E}(Y) - \beta_1 \mathbb{E}(X_2). \quad (18)$$

For example, if $X_2 \sim \mathcal{N}(0, 1)$ and $Y = X_2^2$, then $\beta_1 = 0$ and $\beta_0 = 1$. The best linear predictor is the constant 1, whereas the conditional mean is X_2^2 . Here $u = X_2^2 - 1$ is orthogonal to both 1 and X_2 , but $\mathbb{E}(u | X_2) = X_2^2 - 1$ is not zero almost surely.

Acronyms

LIE Law of Iterated Expectations. 1–3